

Missouri University of Science and Technology

Intro to Data Science
CS 3402 (Spring 2026)

Lecture 1 Introduction to Course

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Self Introduction

Xiaowei Yu

- Assistant Professor
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- https://shawey94.github.io/

Research Interests

- Medical AI, Machine learning and Brain-inspired AI, LLMs

Lecture

- When: Tue. & Thu. 11:00am ~ 12:15pm
- Where: CS 121
- Lecturer: Xiaowei Yu (Office 304)
  xych8@mst.edu
- Office hour: Tue. & Thu. 1:00pm ~ 3:00pm

Study Materials

Require Experiences for Course:

- Mathematics (calculus, statistics)
- Data Structure

Textbook:

- *Introduction to Data Science , A Python Approach to Concepts, Techniques and Applications, Laura Igual, Santi Seguí, 2024.*

*You can get free pdf of the textbook via University Library

Study Materials

Some good books for reading:

- *Machine Learning*, Tom Mitchell.
- *Elements of statistical learning*, Friedman, Hastie, & Tibshirani. *Springer*, 2009.
- *All of statistics*, Wasserman Larry. *Springer*, 2004.

Internet:

- Wiki, LLMs, AI conference papers etc.

Course Organization

Lectures

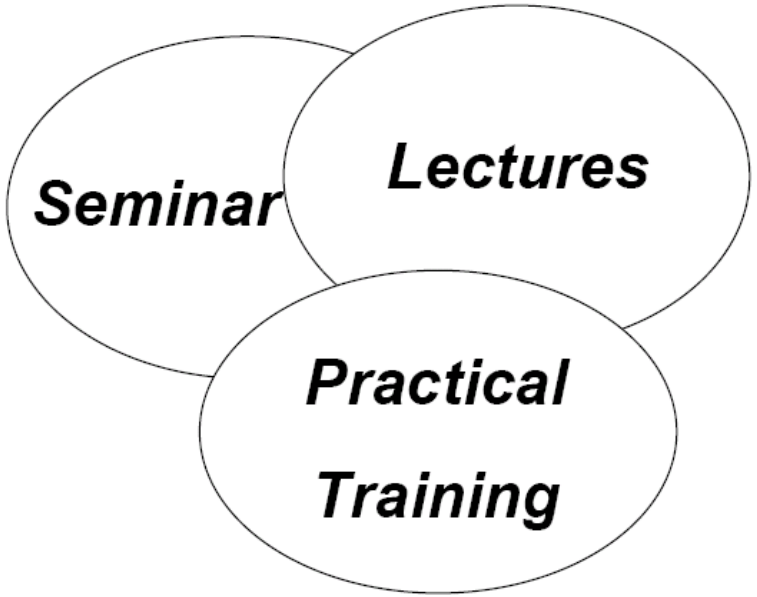
- Intro to data science
- Data science tools
- Machine learning and deep learning models

Practical Training

- Homework
- Quizzes

Seminar

- Project presentation



Course Organization

Something about this course

- You might be familiar with some content (e.g., machine learning models, deep learning models, basics of data science, etc.)
- We need math
- Go deep for some topics, not for all
- AI-heavy

You should:

- **Finish the homework, quizzes independently!**
- **Form a team for group project.**

Course Organization

Grading

- Group project: 30%
- Quizzes: 40%
- Homework: 30%

Quizzes:

- Four quizzes anticipated

Project

- The project will be notified in class and at canvas
- One Programming project

Homework:

- Two-three weeks for each homework
- Three – Four homework anticipated

Group Project

- Get to know your classmates
- Form a team of 2-4 people
- Use AI models (vision/language/multimodal models) to solve a specific application
- Write a report
- Present your project in class

*Details will be introduced later.

Group Project

- Present your project in class
- Presentation ~20 minutes
- Prepare Q&A

*Details will be introduced later.

Course Organization

Grading

Group project:	30%
Quizzes:	40%
Homework:	30%

For example, if you have 80 (project), 90 (homework), 80 (quiz), you final score will be:

final score = 80 * 0.3 + 90 * 0.3 + 80 * 0.4 = 83

The threshold of A will be Avg, the cutoff of B will be Avg – 2*STD.

For example, if the average score of the class (all the students) is 80.5, STD is 4.2, the cutoff of A will be 80.5, and Cutoff B will be 72.1. In this situation, if you final score is 81.7 you will get A, and if your final score is 70, you will get B.

Note: Bonus points are available to improve your final score

Deadlines

- Everything will be submitted through Canvas.
- Due time: 11:59pm
- Late submission: 5-point deduction per hour, till you get 0. 12:01am –5, 1:01am -10, … (The raw score of each submission is 100. So there is no point to submit it after 19 hours).

Regrading

- 7 days after we post scores on Canvas, TA will handle regrade requests. Won't consider it after 7 days.

- If not satisfied with the results, 7 days to request again. Instructor will handle it, and the decision is final.

Your Email

- Make sure your MS&T email account works. We will only contact you by your MS&T email. Check it on a daily basis.

Attendance

- In general, attendance is not required.
- Some content is very important; attendance is strongly encouraged, and bonus points will be awarded for attending certain sessions.
- Notification will be given if the next class includes bonus points for attendance.

Notes

We need a volunteer to present PyTorch/IDEs. (send me you name and MST ID)

Will upload the project/homework to Canvas.

TA:

PyTorch:

IDE:

Python & Numpy:

Academic Integrity

Violations

- Cheating on test/assignment; Plagiarism; Collusion

Can I refer to external materials?

- Yes, but in your homework, source code, and documentation you must explicitly acknowledge the source of information.
- If you copy sentences (completely or partially) from other places, you must enclose them with quotation marks, in addition to provide references to the information source.
- Even if you rephrase, you still need to acknowledge the source.
- If you copy source codes (completely or partially) from other places, you must provide references to the information source.

Academic Integrity

What types of discussions are allowed?

- You can discuss topics related to assignments with your fellow students.
- But you cannot discuss your solutions.
- You must not provide your work (email, hard copy, or in any form) to anyone for any purpose. Following actions are not acceptable:

  "I emailed it to my roommate/friend so that I can submit from their computer, since I couldn't get online from mine."

  "I sent it to my roommate/friend so that I can compile and test my program on their computer, since mine was down."

Academic Integrity

Tutorial: https://grad.mst.edu/faculty/academicintegrity/

More information at: https://grad.mst.edu/faculty/academicintegrity/studentresourcesforacademicintegrity/

The chance of being caught is large; we use tools to diligently check and compare the documents and source codes that you submit to us.

The consequence is certain:

- I will submit the form of "faculty referral of honor code violation" to the university. No exception!
- Academic penalty in the context of this course: 0 on assignment/exam, reduced grade, failing grade of the course
- Penalty by the university: probation, suspension, expulsion, …

AI Tools

- Not allowed for quizzes/exams
- Allowed: learning, debugging, drafting, refactoring
- Disclosure required for assignments/projects
- You must understand and explain everything you submit

Self Introduction (<30 sec)

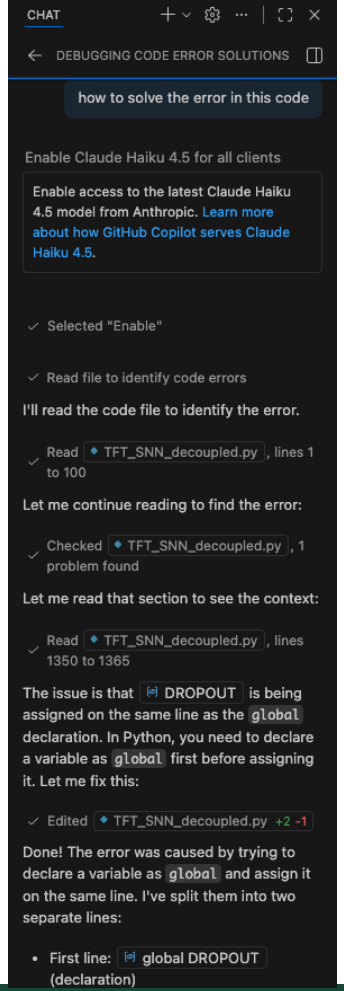
- Name.
- Year.
- What language are you using (Python, Matlab, C, etc.)
- Advisor (if you have).

Working with AI



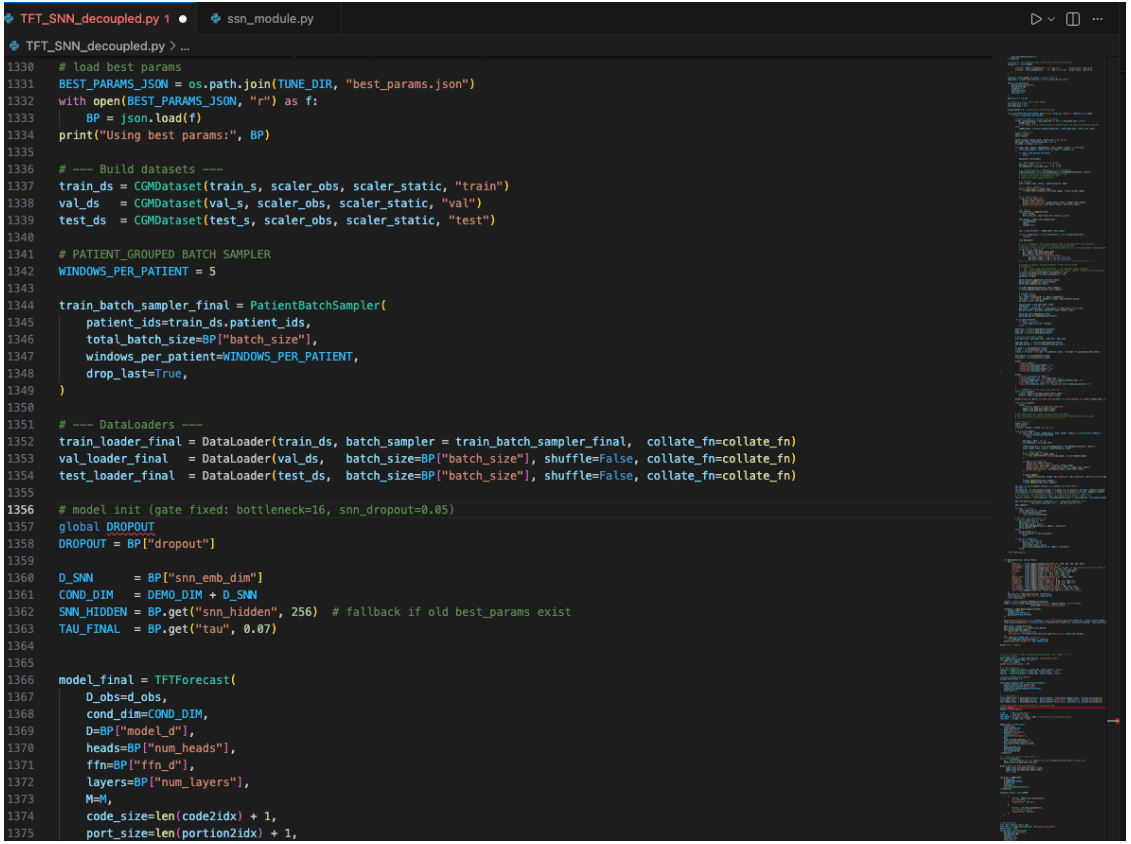
Working with AI

- Identity Bugs
- Provide Solutions
- Save Time
- Improve productivity

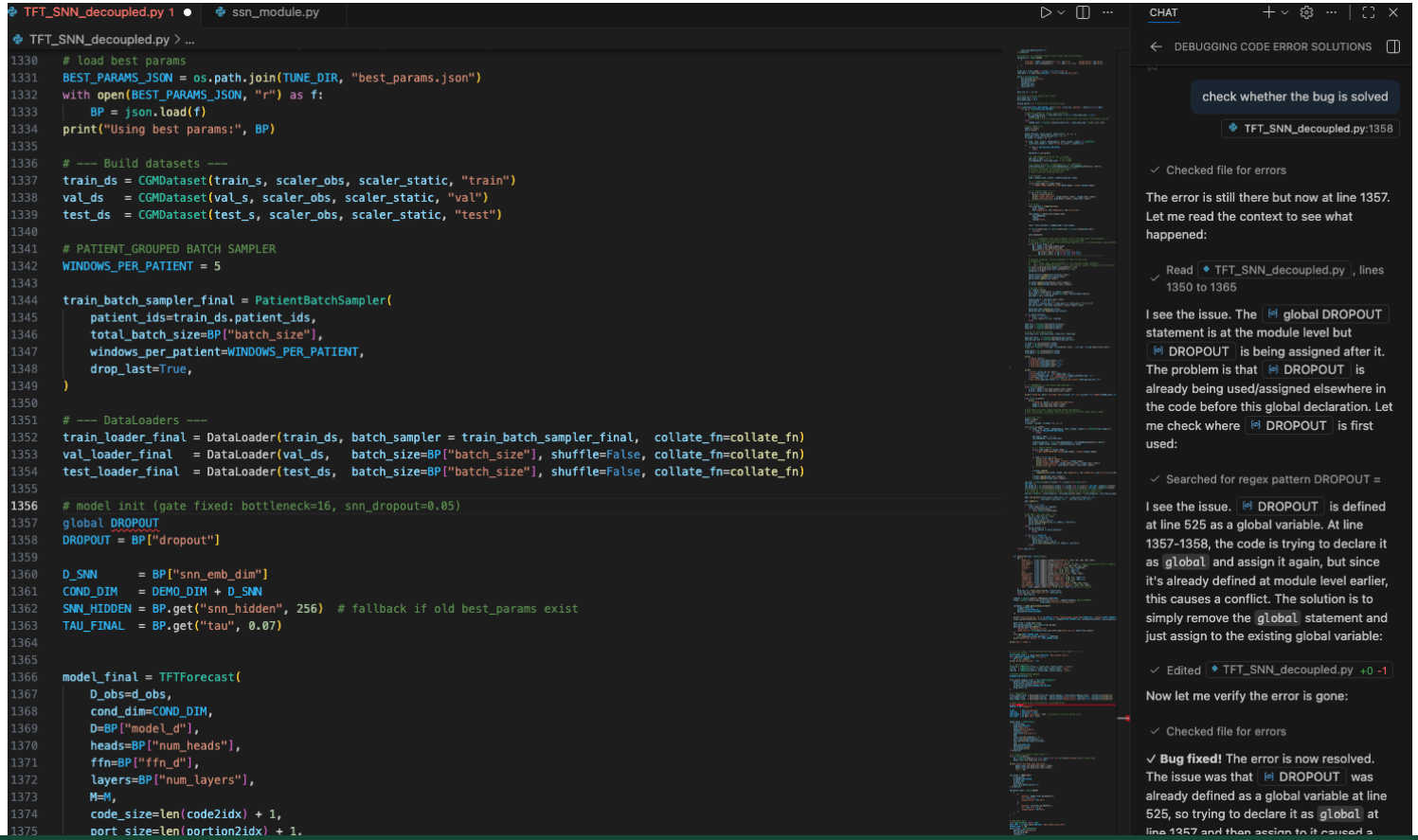


Working with AI

- AI is not perfect
- AI can make mistakes
- Complicated errors need multi-round interactions



Working with AI



Intro to Data Science

Solving problems with data



What makes a good data scientist?



Rethink of Data Science

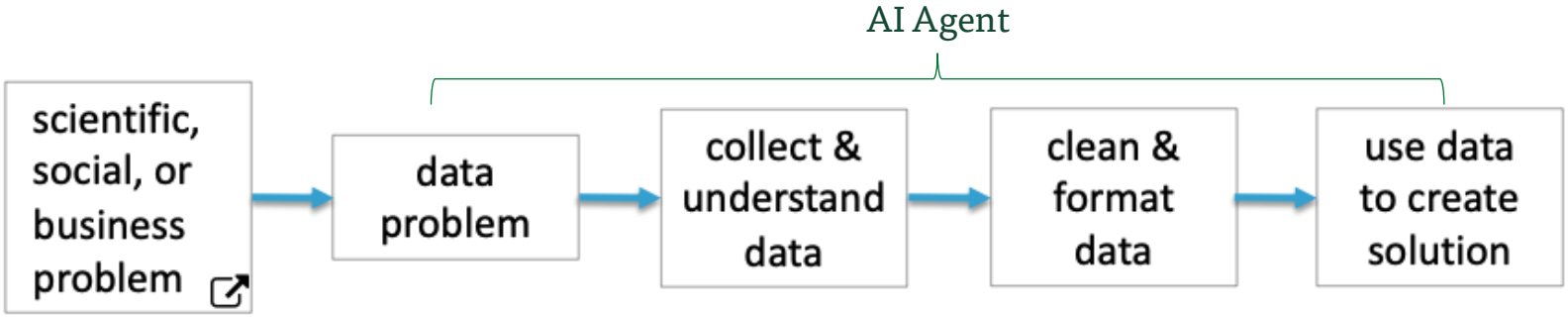
What can AI do in this procedure?



What makes a good data scientist?



AI Agent for Data Science?



What makes a good data scientist?

